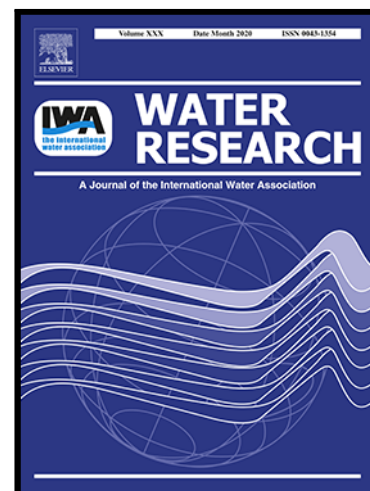


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PII: S0043-1354(21)00678-3
DOI: <https://doi.org/10.1016/j.watres.2021.117480>
Reference: WR 117480



To appear in: *Water Research*

Received date: 13 May 2021
Revised date: 13 July 2021
Accepted date: 26 July 2021

Please cite this article as: Sierra Quinn Sahulka , Bishav Bhattarai , Ananda S. Bhattacharjee , Windy Tanner , Rasool Bux Mahar , Ramesh Goel , Differences in chlorine and peracetic acid disinfections kinetics of *Enterococcus faecalis* and *Escherichia fergusonii* and their susceptible strains based on gene expressions and genomics, *Water Research* (2021), doi: <https://doi.org/10.1016/j.watres.2021.117480>

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Highlights

- Disinfection of antibiotic resistant strains and their susceptible counterparts was studied.
- Meropenem-resistant strain showed less log inactivation compared to susceptible *strain*.
- The resistant *E. faecalis* strain showed similar inactivation compared to susceptible one.
- Meropenem-resistant *E. fergusonii* showed an increase in gene expression of *bla_{NDM-1}* gene to chlorine.
- The meropenem-resistant *E. fergusonii* showed less log inactivation with a marginal significance

Differences in chlorine and peracetic acid disinfections kinetics of *Enterococcus faecalis* and *Escherichia fergusonii* and their susceptible strains based on gene expressions and genomics

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Abstract

This study was conducted to investigate mechanisms of cross-resistance to chlorine and peracetic acid (PAA) disinfectants by antibiotic-resistant bacteria. Our study evaluated chlorine and PAA based disinfection kinetics of erythromycin-resistant *Enterococcus faecalis*, meropenem-resistant *Escherichia fergusonii*, and susceptible strains of these species. Using the integrated second-order disinfectant decay model and first-order Chick-Watson's Law, it was found that the meropenem-resistant *Escherichia fergusonii* strain showed significantly less log inactivation compared to the susceptible *E. fergusonii* strain in response to both chlorine and PAA disinfection (p-value = 0.059, 3.5×10^{-6}). On the other hand, the susceptible *Enterococcus faecalis* strain showed similar log inactivation compared to the erythromycin-resistant strain in response to either treatment (p-value = 0.075, 0.28). Meropenem-

resistant *E. fergusonii* showed an increase in gene expression of New Delhi metallo- β -lactamase (*bla*_{NDM-1}) gene to chlorine, but there was no increase in expression to PAA. Whole genome sequencing (WGS) was then conducted to elucidate the differences in genes among both resistant and susceptible table *E. fergusonii* strains. The average nucleotide identity (ANI) analysis of the draft genomes (>97% similarity) suggests that meropenem-resistant *E. fergusonii* (S1) and meropenem-susceptible *E. fergusonii* (S2) are the same species but different strains. Both strains have the same genes for oxidative stress pathways, oxidative scavenger genes, and nearly 40 different antibiotic efflux pump genes. The chromosomal and plasmid draft genomes of meropenem-resistant and -susceptible *E. fergusonii* strains each have 65 and 52 antibiotic resistance genes, respectively. Of these, the resistant *E. fergusonii* strain harbored the extended-spectrum beta-lactamases *bla*_{CTX-M-15} and *bla*_{TEM-1} genes located on the chromosome, and a *bla*_{TEM-1} gene on the plasmid. The overall findings of this study are significant, as they reveal that antibiotic-resistant and susceptible strains of *E. fergusonii* exhibit different responses towards chlorine and PAA disinfection.

Key words: Antibiotic resistance, disinfection kinetics, expression of genes, peracetic acid, *Enterococcus faecalis*, New Delhi metallo- β -lactamase

1.0 Introduction

Antibiotic resistance poses one of the most serious threats to global human health in the 21st century (Ahmad and Khan 2019, Choudhury et al. 2012, Munita and Arias 2016). Antibiotic-resistant bacteria (ARBs) have evolved to carry antibiotic resistance genes

(ARGs) through mutation, selection, and clonal expansion, especially in environments with strong selective pressure such as hospitals (Palencia and Maria 2018). Resistant infections are typically discovered in nosocomial settings ten years after antibiotics have been introduced to the public (Ventola 2015). Today, even the most powerful antibiotics have proven ineffectual against multidrug-resistant bacteria, such as those carrying the New Delhi metallo- β -lactamase (*bla_{NDM-1}*) gene (Lee et al. 2018, Linciano et al. 2018). Although ARBs are found naturally in the environment, the overuse and misuse of antibiotics in humans, animals, and agriculture has resulted in the proliferation of antibiotic-resistant pathogens worldwide (Levy and Marshall, 2009, Czekalski et al. 2016, Silley et al. 2012). These ARBs and ARGs are now considered Contaminants of Emerging Concern (CECs) in aquatic environments (Sanderson et al. 2016). Unlike other contaminants, ARBs and ARGs are conserved in bacterial populations through horizontal gene transfer (HGT) (Aminov et al. 2011). Therefore, ARBs and free nucleic ARGs often end up in drinking water sources and treatment facilities through hydrologic cycling, where they pose a threat to human health (Chen et al. 2015, Segura et al. 2009).

ARBs and ARGs have been shown to survive disinfection processes in wastewater and drinking water treatment (Yuan et al. 2015, et al. Huang 2011). Traditional chlorine disinfection continues to be the most popular choice for tertiary treatment due to its stability, efficiency, and low cost (EPA 1999). Drinking water treatment facilities (DWTFs) typically add between 0.5 and 2 mg/L of free chlorine to disinfect and combat microbial regrowth in drinking water distribution systems (DWDS) (Sadiq and Rodriguez 2004, CDC 2014). However, the concentration of residual chlorine decreases due to various physico-chemical factors as the water travels downstream in the distribution network (Bertelli 2018). Chlorination also has several drawbacks, namely that it reacts with natural

organic matter to produce numerous disinfection byproducts (DBPs) that have been shown to have carcinogenic and toxic effects (Krishna et al. 2007; Ran and Li, 2013).

Peroxyacetic or peracetic acid (PAA) has been proposed as a green alternative to chlorination because it does not produce DBPs and provides stronger oxidative effects, although it is not as effective as chlorine for residual disinfection in drinking water distribution systems (Zhang et al. 2019). Chlorine bleach simply dissociates into hypochlorite ions in water, while PAA exists in aqueous equilibrium as acetic acid and hydrogen peroxide, producing further oxidative effects (Sharma 2016). Both disinfectants kill pathogens by oxidizing the cell wall and causing lysis, or by diffusing through the cell wall and oxidizing the intracellular material (EPA Chlorine Disinfection 1999, EPA Peracetic Acid Disinfection 2012). Although PAA has been used for many years in European wastewater treatment, its use has not been implemented in the United States and is a virtually nonexistent practice for drinking water treatment due to limitations in cost and operational knowledge (Xue et al. 2017). The Environmental Protection Agency has not yet established guidelines for PAA disinfection in wastewater and drinking water treatment. Furthermore, the efficacy of PAA as a disinfectant for the removal of ARBs and ARGs in wastewater and drinking water treatment is still under investigation.

Recent studies suggest that both chlorine and PAA disinfection in potable water and wastewater treatment can increase the relative abundance of ARGs and ARBs in the final effluent (Liu et al. 2018, DiCesare et al. 2016). Bacteria acquire resistance to multiple environmental stressors via three main mechanisms: a) cross-resistance; b) co-resistance; and c) co-selection (Krige et al. 2009; Pal et al. 2017; Blanco et al., 2016; Bouki et al. 2013, Krige et al. 2009). While it is possible that an oxidative-resistance gene may be co-selected for the same plasmid as ARGs, very little is known about the genetic mechanisms of resistance to oxidative stress

(Mara and Horen 2003). Multidrug efflux pump systems have been strongly implicated in resistance to oxidizing disinfectants (Inatsu 2010, Karumathil et al. 2014). However, results from previous studies show that oxidizing disinfectants increase the gene expression of other classes of ARGs, not just efflux pumps (Karumathil et al. 2014; Templeton et al. 2009, Huang et al. 2013). In this study, we investigated the phenomenon of cross-resistance in antibiotic-resistant *Enterococcus faecalis* and *Escherichia fergusonii* and two corresponding susceptible strains. Certain strains of *Enterococci*, such as *E. faecalis* and *E. coli*, the latter of which is a closely related species to *E. fergusonii*, are pathogens that can cause fatal gastrointestinal diseases when ingested via fecally contaminated drinking water (Byappanahalli et al. 2012, Riley et al. 2011). The *E. fergusonii* strains that cause infections are also known to be highly resistant to multiple antibiotics such as chloramphenicols, ampicillin, and gentamycin (Maheux et al. 2014). For this study, these two species were selected as model waterborne pathogens, and they were chosen based on their ability to phenotypically resist completely unrelated classes of antibiotics that do not involve efflux pumps, in order to explore other potential mechanisms of cross-resistance.

Therefore, the overall objective of this study was to compare the disinfection kinetics of antibiotic-resistant *E. faecalis* and *E. fergusonii* and their corresponding susceptible strains. **The objective also included was to investigate a potential mechanism for cross-resistance** to antibiotics and oxidizing agents. We chose two bacteria (e.g., *E. fergusonii* and *E. faecalis*) from different phyla with two completely disparate modes of antibiotic resistance that does not include efflux pumps to see if antibiotic resistance genes themselves were responsible for cross resistance, or whether other mechanisms of resistance to oxidizing disinfectants are possible. It was hypothesized that antibiotic-resistant pathogens would show less log inactivation and greater expression of ARGs in response to chlorine and PAA disinfection. It was also hypothesized that the draft genomes (bacterial chromosome and plasmid) of meropenem-

resistant *E. fergusonii* and susceptible *E. fergusonii* strains would show differences in the number and diversity of ARGs. Additionally, resistance mediated by erythromycin resistance methylase and encoded by *erm* genes *ermA* and *ermB*, and, and additional active drug efflux gene *mefA* (Kataja et al., 2000). Hence, in erythromycin *E. faecalis*, the expression of these functional genes was also studied in response to chlorine and PAA disinfection. Likewise, *E. fergusonii* harbors the new Delhi metallo- β -lactamase *bla_{NDM-1}* gene which offers resistance to the broad spectrum Carbapenem group of antibiotics (Ashcroft et al., 2019). Hence, the expression of this gene was studied in response to chlorine and PAA disinfection.

2.0 Methods

2.1 Isolation and Characterization of Antibiotic-Resistant Pathogens

The antibiotic-resistant pathogens used in this study were isolated from agricultural and clinical samples. The meropenem-resistant *E. fergusonii* was isolated in Dr. Ramesh Goel's laboratory at the University of Utah in a previous study, was phenotypically determined to be an *Escherichia coli* strain using plating methods, and genotypically determined to carry *bla_{NDM-1}* using PCR (Tanner et al. 2017). The erythromycin-resistant *E. faecalis* was isolated by diluting one gram of swine manure sample in 0.85% w/v saline solution. A minimum inhibitory concentration (MIC) test was performed using 98% mg/g erythromycin, serially diluted from 1 ug/mL to 512 ug/mL. The diluted manure sample was four-quadrant streaked on CHROMagar Orientation chromogenic Mueller Hinton agar (MHA) containing erythromycin and incubated at 37 °C for 24 hours. Based on distinct colony morphology and color, *E. faecalis* colonies were picked from the 512 ug/mL erythromycin containing plate and cultured in Mueller Hinton broth (MHB). The individual

isolates were subjected to a three-level subculture screening on MHA containing 64 ug/mL erythromycin to confirm antibiotic resistance, and to negate the possibility of false positive results.

Both the susceptible *E. faecalis* and *E. fergusonii* strains were obtained from a colleague in Civil and Environmental Engineering Department at the University of Utah, and susceptibility was tested by plating the culture on 64 mg/L erythromycin- and 4 mg/L meropenem-containing plates, subsequently showing no growth. In order to estimate an initial bacterial concentration of 10^8 CFU/mL for disinfection, growth curves were developed for each strain by measuring the optical density of each bacterial culture at 600nm in a spectrophotometer, and the time was recorded when each strain reached mid-log phase growth. Meropenem-resistant and susceptible *E. fergusonii* were cultured in MHB, while erythromycin-resistant and susceptible *E. faecalis* showed optimal growth in Tryptic Soy broth (TSB). The isolated bacterial cultures were examined microscopically using an epifluorescence microscope (BX 51 Olympus, Japan) and 4',6-diamidino-2-phenylindole fluorescent staining, to confirm a pure culture and correct morphology had been obtained, before being stored in 25% sterile glycerol. Detection of respective antibiotic resistance genes, e.g., for *E. faecalis*, the erythromycin resistance was tested with *ermA*, *ermB*, and *mefA* genes and for *E. fergusonii*, the *bla_{NDM-1}* gene, was carried out using primer sets and PCR conditions as detailed in Li et al., 2010 and Poirel, 2011 respectively.

2.2 Chlorine and PAA Inactivation of Antibiotic-Resistant Pathogens

For the chlorine inactivation tests, a known volume of sodium hypochlorite (7.85% w/v) was added to 100 mL of sterilized 0.85% saline solution for a 50 mg/L stock solution, and diluted to produce concentrations of 2, 4, 8, and 16 mg/L residual free

chlorine. Free chlorine concentrations were confirmed using the HACH free chlorine DPD method 10069 (HACH, USA) and a HACH DR1900 spectrophotometer (HACH, USA). For the PAA inactivation tests, a known volume of PAA (35% w/v) was added to 100 mL of sterilized 0.85% saline solution for a 6 mg/L stock solution and diluted to produce concentrations of 0.5, 1, 2, and 5 mg/L residual PAA. The PAA concentrations were confirmed using the HACH PAA DPD method 10290 (HACH, USA) and DR1900 spectrophotometer. Measurement of PAA and H₂O₂ concentration is adapted from DPD method for chlorine (Dominguez-Henao et al. 2017, HACH, USA).

Four complete-mix batch reactors assays, representing each disinfectant concentration, were constructed in 100 mL flasks on stir plates set to 400 RPM, and strains were tested individually. The experiments were conducted at room temperature (20 °C) and neutral pH (7.5). By taking a culture sample at mid-log growth phase, approximately one mL of 10⁸ CFU/mL culture was added to each 100 mL assay with a stir bar to produce an approximate final concentration of 10⁶ CFU/mL. Subsequently, one mL sample was taken at time 0, 5, 15, 25, and 35 minutes. A known volume of 0.02 M sodium thiosulfate was added to each sample to quench both residual chlorine and PAA (Eramo et al. 2017). The sample was then serially diluted from 10⁶ CFU/mL to approximately one CFU/mL, filtered through a 0.22 µm filter (Which Company, USA), and placed on a 47 mm petri dish containing a selective chromogenic agar. For *E. faecalis*, modified Difco MEI agar was used and the strain was incubated at 42 °C for 24 hours. For *E. fergusonii*, modified Difco MTec agar was used and the strain was first incubated at 37 °C for two hours, followed by incubation at 42 °C for 22 hours. Colony forming units were counted on the filter paper for each dilution, and the original bacterial concentration was

calculated for each time and concentration. The inactivation experiments were performed as technical triplicates in order to calculate the average and standard deviation of bacterial concentrations over time for each treatment.

In order to calculate true contact time, the instantaneous demand (in a time scale of minutes), decay, and residual concentrations of free chlorine and PAA were measured in separate experiments under the exact same conditions (McClellan et al. 2000, Metcalf and Eddy, 2014). Five- and ten-mL aliquots of free chlorine and PAA, respectively, were taken at time 0, 5, 15, 25, and 35 minutes. They were immediately analyzed spectrophotometrically using HACH methods 10069 and 10209, and the average residual concentrations obtained from technical triplicates were used to calculate integrated second-order demand and decay model for contact time (C_T) values (Gyurek et al. 1998, Sivaganesan et al. 2008). Additional complete-mix batch reactor assay experiments were performed to confirm that the instantaneous demand was mainly exerted by each of the culturing medias, and that the observed residual disinfectant concentrations were the same between resistant and susceptible strains.

2.3 Disinfection Demand and Kinetics

The number of surviving bacteria at each sampling time point was calculated after accounting for the dilution ratio and the aliquot volume used during each experiment. In conditions where disinfectant demand and decay are negligible, the disinfectant concentration is assumed to stay constant over the course of the contact time, and zero order kinetics may be used to determine C_T values. However, in this study, all residual disinfectant experiments for both chlorine and PAA exhibited both instantaneous demand and disinfection decay over time; therefore, a second-order model of disinfectant demand and decay was used to determine an

integrated estimate of contact time over the course of the experiment. The differential form of this second-order model is given by Equation 1, and the integrated form of Equation 1 gives the residual disinfectant concentration at any time in Equation 2. In Equation 1, C is equal to the initial concentration of disinfectant, C^* is equal to the amount of disinfectant demand at any given time, and $-k_2$ is a disinfectant decay constant, the units of which are $(1/t)$. In Equation 2, $(C)_t$ is equal to the residual disinfectant concentration at any given time t , $(C)_0$ is equal to the initial concentration of disinfectant, and r and $-k_2$ are disinfectant demand and decay constants, respectively. The constant r equals $(1 - C^*)$ at any given time t , the units of which are (mg/L) .

$$\frac{dC}{dt} = -k_2 C(C - C^*) \quad (1)$$

$$(C)_t = (C)_0 \cdot \frac{(1 - r)}{1 - r \cdot e^{-k_2 \cdot (1-r) \cdot t}} \quad (2)$$

Using the experimental residual concentrations of chlorine and PAA at each time point and a house-written Python script, iterations were performed to calculate r and $-k_2$ constants for each strain and corresponding culturing media (Sivaganesan et al. 2008). An integrated estimate of true contact time (denoted as C_t in Equation 3) was then calculated as the product of the initial disinfectant concentration and the integrated form of Equation 2, as described by Equations 3, 4, and 5 (Sivaganesan et al. 2008).

$$\log(N_t/N_0) = -K_s C_t \quad (3)$$

$$\log(N_t/N_0) = -K_s \cdot (C)_0 \times \int_0^t \frac{(1 - r)}{1 - r \cdot e^{-k_2 \cdot (1-r) \cdot x}} dx \quad (4)$$

$$\log(N_t/N_0) = -K_s \cdot (C)_0 \cdot \{(1-r) \cdot t + [(1/k_2) \times \ln(1 - r \cdot e^{-k_2 \cdot (1-r) \cdot t})/(1-r)]\} \quad (5)$$

The initial disinfectant concentrations that produced a gradual decline in bacterial counts over time were 4 mg/L and 8 mg/L for chlorine and 1 mg/L and 2 mg/L for PAA (Figures 1a-d and 2a-d). Minimal deviations from first-order kinetics were observed for these treatments, which indicate lagging or tailing responses to disinfection, therefore a first-order model of Chick-Watson's Law was used to calculate the kinetic constant for each strain (Gyurek et al. 1998). Since Chick-Watson's Law assumes that different manifestations of C_T for a certain disinfectant will result in the same magnitude of log inactivation of a particular microorganism, the bacterial counts from the aforementioned experiments were combined into one dataset to examine the **overall** disinfection kinetics of each bacterial strain and disinfectant type. It should be noted that this assumption does not often hold in real world conditions, even in controlled experiments, and that different initial concentrations of disinfectant usually produces dose-dependent responses in log inactivation (Peleg 2020). However, these responses are minimized in large datasets, as seen in Sivaganesan et al. 2008. Therefore, total log inactivation values ($\log(N_t/N_0)$) were calculated by taking the log base 10 of the number of surviving bacteria divided by the initial number of bacteria. The log inactivation values were then plotted against their corresponding integrated C_T values, and the slope of each resulting scatter plot, which is equal to the kinetic constant K_s , was determined graphically using linear regression analysis in RStudio (Figures 3a-d and Table 1).

2.4 Expressions of *ermB* and *bla*_{NDM-1} Antibiotic Resistance Genes

Inactivation experiments were repeated on the antibiotic-resistant bacterial strains exactly as detailed earlier to capture the expression of *ermB* and *bla*_{NDM-1} genes during disinfection. For the chlorination response test, five 100 mL flasks containing 10⁶ CFU/mL of either erythromycin-resistant *E. faecalis* or meropenem-resistant *E. fergusonii* were treated with 0 mg/L and 8 mg/L free chlorine. Each of the five flasks represents a 100 mL aliquot to be filtered at time 0, 5, 15, 25, and 35 minutes, respectively. For the PAA response test, similar batch tests were performed using 0 mg/L and 2 mg/L PAA for the same time frame. Disinfection concentrations were chosen based on previous results which produced a significant difference in disinfection kinetics between the meropenem-resistant and susceptible *E. fergusonii*. Each of the 100 mL batch cultures was placed on 0.22 µm filters (Millipore, USA) at its respective time, and preserved in RNAlater (Thermo Fisher Scientific, USA) for further processing. RNA was extracted using the PureLink RNA Mini Kit (Invitrogen Corporation, Carlsbad, CA). After the extraction of RNA, reverse transcription was conducted to convert approximately 2500 ng RNA to cDNA in each sample using the SuperScript VILO cDNA synthesis kit (Life Technology, NY).

A QuantStudio 3 Real-Time PCR system (Thermo Fisher Scientific, USA) was used to analyze the expression of *ermB* and *bla*_{NDM-1} genes with the delta-delta-C_T method, which was developed by Schmittgen and Livak (2008). For each sample, the abundance of *bla*_{NDM-1} and *ermB* gene transcripts, whose primer sequences are specified in the aforementioned sections, were first quantified in the machine using standard qPCR measurements, but then was normalized by the abundance of 16S rRNA gene

transcripts present (the reference or “housekeeping” gene). The results were then expressed as fold change relative to the ARG gene transcript number measured at time zero. Therefore, the units of the relative quantity of the delta-delta- C_T are: ARG gene copy number at time t divided by the product of the 16S rRNA gene copy number at time t and the ARG gene copy number at time zero, which is unitless. A master mix was created for each primer set, in which 10 μ L AB Power SYBR® Green Master Mix, 1 μ L of each primer (10 mM), 1 μ L of 20 ng/ μ L DNA template and nuclease free water were added to make a total volume of 20 μ L. The qPCR process was modified using the VeriFlex temperature control technology during the annealing process. It was followed by 2 min at 2 min at 50 °C, 5 min at 94°C; then 40 cycles consisting of 60 s at 94 °C (denaturation), 90 s at 57°C for the *bla*_{NDM-1} gene, 52 °C for the *ermB* gene, and 60°C for the 16S rRNA gene (annealing using VeriFlex mode), 90 s at 72°C (elongation) and a final cycle consisting 10 min at 72°C.

2.5 Whole-Genome Sequencing and Draft Genome Analysis

Whole-genome sequencing (WGS) was performed to examine other genetic pathways potentially responsible for cross-resistance exhibited by the meropenem-resistant *E. fergusonii* strain. Each resistant and susceptible strain was cultured overnight in MHB at 37 °C from a glycerol stock stored at -80 °C. The resistant strain was then four-quadrant streaked on an mTEC plate containing 4 ug/mL meropenem, while the susceptible strain was streaked onto mTEC plates with and without 4 ug/mL meropenem to reaffirm susceptibility. Single colonies were picked and the isolates were subjected to a three-level subculture screening on their respective plates. A final culture of each was prepared and the biomass pelleted from a 10 mL aliquot. The genomic DNA (gDNA)

was extracted from the biomass pellet according to Griffiths et al. 2000. Total gDNA was sent for sequencing to the Huntsman Cancer Institute High-Throughput Genomics center at the University of Utah. A gDNA library was prepared using a Nextera DNA Flex Library Prep Kit (Illumina Inc, CA, USA) and then sequenced on MiSeq Benchtop sequencer (Illumina Inc, USA). The adapters from the 150 bp paired-end raw reads were removed using TagCleaner v. 0.16 (Schneider et al. 2010). The quality filtering of sequence reads for a mean Phred quality score greater than 25 was performed using Prinseq-Lite v. 0.20.4 (Schneider and Edwards 2011), resulting in less than 5% loss of total reads. The CLC Genomics Workbench (QIAGEN, USA) was used to perform *de novo* assembly of each genome using various parameters. A word size of 64 and bubble size of 200 was chosen to assemble the final genomes since these parameters resulted the highest N50 score, or the median proportion of contigs consisting of the greatest number of base pairs, and lowest number of contigs, or most complete genomes (Alhakami et al. 2017). Contigs less than 1,000 bps were then removed using an in-house developed python script. The summary of sequenced genomes are shown in Table 2.

2.6 Gene Synteny Analysis of Bacterial Chromosomal and Plasmids

The alignment blocks of the two genomes (chromosomes and plasmids) were assessed with synteny analysis. In brief, the quality of the assembled genomes was evaluated using CheckM (Park et al., 2015). The draft genomes (chromosomes and plasmids) assemblies were > 99% complete with < 0.5% contamination. The resulting meropenem-resistant (S1) and susceptible (S2) *E. fergusonii* genomes were determined to be high-quality assembled genomes, the summary statistics of which are shown in Table 3 (Bowers et al., 2017). Phylogenetic analysis of the two *E. fergusonii* draft chromosomal genomes was performed using with the recovered 16S rRNA gene. National Center for Biotechnology Information (NCBI), nucleotide Basic Local Alignment Search Tool

(BLASTn; Altschul et al., 1990; Zhang et al., 2000) analysis with 16S ribosomal RNA (Bacteria and Archaea) was used for finding the microorganism that is phylogenetically closest to the meropenem-resistant (S1) and susceptible (S2) *E. fergusonii* genomes. *E. fergusonii* ATCC 35469 was assessed to be the closest known microorganism to both *E. fergusonii* strains. The reorder draft bacterial chromosome and plasmid genomes of meropenem-resistant (S1) and susceptible (S2) *E. fergusonii* with the lowest locally collinear blocks (LCBs) was chosen for downstream analysis (Rissman et al., 2009). The relative ordering of the meropenem-resistant (S1) and susceptible (S2) *E. fergusonii* draft genome (chromosome and plasmids) scaffolds was performed based on the *E. fergusonii* genome using Mauve Aligner (Rissman et al., 2009) with default parameters.

Genome annotation of the assembled reordered draft bacterial chromosome and plasmid genomes was performed using PROKKA (Seeman et al., 2014). Easyfig v 2.2.2 was used for synteny analysis and visualization (Sullivan et al., 2011). Post gene prediction and annotation, a resistance gene identifier (RGI) analysis within the Comprehensive Antibiotic Resistance Database (CARD, Alcock et al. 2020) was used to identify gene that could confer antibiotic resistance to the meropenem-resistant (S1) and susceptible (S2) *E. fergusonii* strains. A BLASTn analysis was performed, with the draft bacterial chromosome and plasmid genome of meropenem-resistant (S1) *E. fergusonii* as query and susceptible (S2) *E. fergusonii* as subject, to evaluate the chromosomal and plasmid genome alignments. The BLASTn results of the draft genome (chromosomal and plasmid) alignments was visualized. The average nucleotide identity (ANI) between the meropenem-resistant (S1) and susceptible (S2) *E. fergusonii* genome (chromosomal and plasmid) sequences was calculated using OrthoANiU algorithm (Yoon et al., 2017), and ANI calculator (Gorris et al, 2007). The

OrthoANlu(%) and ANI(%) are 97.42 and 97.56, respectively. The ANI (%) > 97% suggests meropenem-resistant (S1) and susceptible (S2) *E. fergusonii* are bacterial strains for same species (Gorris et al., 2007).

2.7 Statistical Analysis

The significance of difference in disinfection kinetics between antibiotic-resistant and susceptible strains was assessed by performing an Analysis of Covariance (ANCOVA) using R statistical software. The significance of gene expression between the control culture and the culture treated with a disinfectant after 35 minutes was assessed by performing a paired two-tailed t-test using R statistical software. Considering a small data size in gene expression analysis, a non-parametric Wilcoxon rank sum test was also conducted. All of the results were considered significant with a p-value less than 0.05. However, some uncertainty has recently been raised around the meaningfulness of results after the imposition of a strict cut-off p-value of 0.05, therefore in this study a p-value of less than 0.6 was also considered significant (Andrade 2019, Grabowski 2016).

3.0 Results

3.1 Chlorine and PAA Disinfection Kinetics and Expression of *ermB* and *bla_{NDM-1}* Genes

It was expected that the growth media would exert chlorine demand. Hence, although breakpoint chlorination experiments were not conducted, the initial amount of sodium hypochlorite and PAA needed to maintain residual disinfectant concentrations were pre-determined using batch experiments. Integrated second-order models of free chlorine and PAA demand and decay for each media

and species are shown in Supplementary Figure S1a-d. The instantaneous demand and disinfectant decay were similar between species and strains, and minimally affected the integrated second-order model C_T values. However, demand and decay differed by the disinfectant that was used. Additionally, most of the demand was exerted by either the MHB or TSB culture media. Based on these experiments, for initial total chlorine concentrations ranging from 16 mg/L to 2 mg/L, residual total free chlorine concentration ranged from 1.9 ± 0.82 mg/L to 0.1 ± 0.05 mg/L after 5 minutes of reaction time. Likewise, for initial PAA concentrations of 5 mg/L to 0.5 mg/L, instantaneous demand resulted in measured concentrations ranged from 2.0 ± 0.58 mg/L to 0.1 ± 0.05 mg/L after 5 minutes. As indicated from the disinfection data, the residual concentrations (e.g. break point chlorination) obtained were sufficient to cause bacterial inactivation. The observed instantaneous disinfectant demand and disinfectant decay supported the use of a second-order model to determine C_T values.

The overall colony counts during chlorine disinfection for both pairs of resistant and susceptible strains of *Enterococcus faecalis* and *Escherichia fergusonii* at different free chlorine concentrations are shown in Figures 1a-d. Figures 1a&b show chlorine inactivation results for meropenem-resistant *E. fergusonii* and susceptible *E. fergusonii*. Likewise, Figures 1c&d show chlorine inactivation results for erythromycin-resistant *E. faecalis* and susceptible *E. faecalis*. In cases where initial chlorine concentration was 2 mg/L, there was little to no reduction in bacteria counts for all strains tested due to low residual free chlorine concentration. Conversely, in cases where initial chlorine concentration was 16 mg/L, there was an immediate inactivation of bacteria and subsequently no growth after five minutes. All strains showed gradual log inactivation over time at 4 mg/L and 8 mg/L initial free chlorine (Figure 1). Therefore, the disinfection kinetics values for resistant and susceptible *E. faecalis* and *E. fergusonii* were

calculated by combining the log inactivation and C_T values for 4 mg/L and 8 mg/L initial free chlorine concentrations (Sivaganesan et al. 2008). Changes in bacterial concentration in response to chlorine dose over time was calculated using the integrated form of the second-order disinfectant demand model and first-order Chick-Watson's Law, and was then statistically analyzed (Sivaganesan et al. 2008, Gyurek et al. 1998). The results of the linear regression analysis on the integrated second order C_t and log inactivation values for chlorine disinfection kinetics are summarized in Table 1.

The overall colony counts during PAA disinfection for both pairs of resistant and susceptible strains at all PAA concentrations are shown in Figures 2a-d. Much like disinfection by chlorine, in cases where the initial PAA concentration was 0.5 mg/L, there was only a one log inactivation in bacteria counts for all strains. In cases where residual PAA concentration was 5 mg/L, all strains were inactivated by 5 minutes. All strains showed gradual log inactivation over time at 1 and 2 mg/L initial PAA. Therefore, the disinfection kinetics values for erythromycin-resistant and susceptible *E. faecalis* and *E. fergusonii* were calculated by combining the log inactivation and C_T values for 1 mg/L and 2 mg/L initial PAA concentrations. The results of the linear regression for PAA disinfection kinetics are shown in Table 1.

The results of the ANCOVA analysis for chlorine disinfection are shown in Figures 3a and 3b for *E. faecalis* and *E. fergusonii*, respectively. The erythromycin-susceptible *E. faecalis* showed slightly less log inactivation than the resistant strain, and the disinfection kinetic constants K_S , as represented by the slopes of the linear regression, were not significantly different (p-value = 0.0750) (Figure 3a). The meropenem-resistant *E. fergusonii* exhibited less log inactivation than the susceptible strain (p-value = 0.0590), and there was a significant difference in disinfection kinetics between the resistant and susceptible strains (Figure 3b). The

ANCOVA analysis of PAA disinfection kinetics for resistant and susceptible *E. faecalis* and *E. fergusonii* are shown in Figures 3c and 3d respectively. Figure 3c shows that erythromycin-resistant and susceptible *E. faecalis* did not show a significant difference in log inactivation (p-value = 0.279), while Figure 3d shows the meropenem-resistant *E. fergusonii* showed much less log inactivation than the susceptible *E. fergusonii*, with significantly different disinfection kinetics constants (p-value = 3.5×10^{-6}). Overall, greater C_T values were required to produce the same log inactivation in *E. faecalis* than *E. fergusonii*, and the initial concentration and integrated C_T values for PAA disinfection were lower than required to produce the same log inactivation by chlorine disinfection (Figures 3a-d).

The expressions of *ermB* and *bla_{NDM-1}* gene encoding New Delhi metallo- β -lactamase in response to chlorine and PAA disinfection are shown in Figures 4a-d in terms of fold change (relative abundance). The relative gene expression in *E. faecalis* for *ermB* between 0 mg/L and 8 mg/L free chlorine after 35 minutes was low overall and not significant (p-value = 0.414) (Figure 4a). Additionally, apart from a paired t-test, a non-parametric Wilcoxon rank sum test was also run and this enabled similar statistics. *Bla_{NDM-1}* encoding New Delhi metallo- β -lactamase in *E. fergusonii* was expressed at a much higher relative abundance in response to chlorine, with nearly 200 gene copies after 35 minutes, but not significantly due to the large standard deviation (p=0.0790) (Figure 4b). The gene expression for the control *ermB* strain mixed with 0 mg/L PAA was significantly higher than the gene expression for *ermB* with 2 mg/L PAA after 35 minutes (p-value = 0.0547) (Figure 4c). This trend was also observed, but not significantly, for gene expression of *bla_{NDM-1}* in *E. fergusonii*, which was expressed slightly higher with 0 mg/L PAA (p-value = 0.152) (Figure 4d).

3.2 Draft Genomes (Chromosomal and Plasmid) and Differences in Disinfection Resistance

Due to the difference in disinfection kinetics exhibited by meropenem-resistant and susceptible *E. fergusonii* strains, as well as the inconclusive results of the gene expression experiments, whole genome sequencing (WGS) was employed to understand the differences in the bacterial chromosomal and plasmid genomes. WGS, assembly, gene prediction, and annotation of predicted genes with special focus on genes that could confer antibiotic resistance was performed on both the meropenem-resistant and susceptible *E. fergusonii* strains. The resulting statistics of the genome sequencing, assembly, and annotation are summarized in Tables 2 and 3. The OrthoANIu value was 97.42% and the average aligned length was 3,158,597 bp. Analysis of each draft chromosomal and plasmid genomes of the meropenem-resistant and susceptible *E. fergusonii* strains using PROKKA revealed a difference in the number of antibiotic resistant genes (ARGs) between the chromosomes and plasmids of the two strains, while maintaining overall similar distribution of ARG types. 20 genes (with 100% similarity) and 45 genes (> 94 % similarity) were identified for meropenem-resistant (S1) *E. fergusonii* chromosome (Figure 5a). A total of 41 genes (with 100% similarity) and 11 genes (<100 % but > 94% similarity) were identified for the susceptible (S2) *E. fergusonii* chromosome (Figure 5a). In addition, two ARGs were identified on plasmid of the meropenem-resistant (S1) *E. fergusonii* chromosome with 100% similarity. However, none were identified for the susceptible (S2) *E. fergusonii* plasmid.

The RGI analysis of the Comprehensive Antibiotic Resistance Database resulted in slightly different findings CARD revealed that the resistant strain contained a total of 74 perfect and strict hits, while the susceptible strain contained only 61 total ARG hits. In total, both strains shared 44 efflux pump genes, 11 target site alteration genes, four antibiotic inactivation genes, and two reduced permeability related ARGs (Figure 5b). The susceptible strain possessed only three unique ARGs, EF-Tu, mdtM, and mphB, only one

of which is an efflux pump, while the resistant strain possessed 15 unique ARGs. Of these 15 unique ARGs, seven of them were related to target site alteration, two were related to extended-spectrum beta-lactamases, two were related to antibiotic inactivation, and three are related to efflux pump systems (Figure 5b). These results showed that the meropenem-resistant *E. fergusonii* carried the *bla*_{NDM-1} gene, but confirmed the presence of other extended-spectrum beta-lactamase genes (ESBLs), including *bla*_{TEM-1} and *bla*_{CTX-M-15}, and two additional antibiotic inactivation resistance genes, BRP(MBL) and *aadA2* (Figure 5b).

The PROKKA analysis also showed that the chromosomes and plasmids of meropenem-resistant (S1) and susceptible (S2) *E. fergusonii* both contained oxidative stress response genes (*soxR* and *soxS*) and peroxide scavenger genes (*katE*, *katG*, *ahpF*, and *ahpC*). A synteny analysis between both bacterial chromosome genomes is shown in Supplementary Figure S2. Additionally, a synteny diagram of plasmid sequence and annotated genes is presented in supplementary figure S3. To acknowledge the limitations of this study: First, both resistant and susceptible strains of *E. fergusonii* were same species based on average nucleotide frequency comparison (>97%) but not necessarily isogenic strains. Nevertheless, genome wide comparison did show many similarities in key functional genes but differences in antibiotic resistance genes. Second, whole-genome transcriptomics, rather than whole-genome sequencing, may have provided a clearer picture of the functionality of resistance mechanisms involved during oxidative stress.

4.0 Discussion

4.1 Cross-Resistance to Antibiotics and Oxidizing Disinfectants

In this study, the differences in disinfection kinetics were assessed for two antibiotic-resistant strains and their susceptible counterparts, as well as the gene expression of their relevant ARGs, in response to chlorine and PAA disinfection. Although it was hypothesized that both antibiotic-resistant strains would show less log inactivation than the susceptible bacteria, only the *E. fergusonii* carrying *bla*_{NDM-1} showed significant differences in disinfection kinetics to both chlorine and PAA treatment. Although first-order Chick Watson's disinfection kinetics was observed with minimal lagging or tailing, when applying the second-order disinfectant demand and decay model according to Sivagansan et al. 2008, there were observed dose-dependent responses among the data at different initial concentrations of disinfectant. Some studies have shown that Chick-Watson's Law of first-order disinfection kinetics do not produce interchangeable C_T values, and that the simplicity of the Chick-Watson's Law concept has certain limitations due to the irregularity and irreproducibility of stochastic models to measure a volatile chemical like sodium hypochlorite (Peleg 2020). However, for the sake of utility, the C_T concept continues to be the most universally used mode of capturing log inactivation and standardization of disinfection kinetics across the country.

Surprisingly, the expression of the *bla*_{NDM-1} gene encoding New Delhi metallo- β -lactamase by *E. fergusonii* increased in response to chlorination, as the relative abundance of *bla*_{NDM-1} encoding New Delhi metallo- β -lactamase was higher than the control after 35 minutes of exposure to the disinfectant. This may indicate that the majority of surviving bacteria are expressing the *bla*_{NDM-1} gene encoding New Delhi metallo- β -lactamase, relative to the microbial community at time zero and those in the control group. It is also possible that chlorine disinfection may upregulate the expression of the *bla*_{NDM-1} gene overall. However, the expression of their respective ARGs by both *E. faecalis* and *E. fergusonii* was higher in the control of PAA after 35 minutes, as was expected, which

indicates that the exposure to PAA may downregulate the expression of *bla*_{NDM-1} gene in *E. fergusonii*. Previous gene expression studies show that PAA may upregulate oxidative stress and membrane transport genes in some strains of *Mycobacterium*, but downregulate DNA repair and oxidative stress genes in others, signifying the complexity of these response systems (Nde et al. 2011, Chang et al. 2006, Wong et al. 2018). Hou et al., 2019 show that chlorine injures unregulated efflux pumps in *Pseudomonas aeruginosa*. The expressions of *bla*_{NDM-1} at a higher disinfectant dose is intriguing because one would expect most of the bacteria succumb to disinfection, as observed for *ermB*-carrying *E. faecalis* in response to chlorine, or to thrive in the absence of a disinfectant, as observed for the controls. Some bacteria tend to over express functional genes under external stresses (Wu et al., 2016; Park and Ely, 2009) and this could be the possibility for the tested bacteria in this research. Other potential reasoning could be that disinfectant injured bacteria tend to offer a greater resistance to antibiotics and hence tend to express more antibiotic resistance gene as demonstrated by Hou et al., 2019 for *Pseudomonas aeruginosa*. These conflicting results indicate that additional mechanisms might be responsible for the observed resistance to oxidative disinfectants exhibited by the meropenem-resistant *E. fergusonii* compared to its susceptible counterpart.

4.2 Mechanism of Antibiotic Resistance by Erythromycin-Resistant *E. faecalis*

The results show that erythromycin-resistant *E. faecalis* strain exhibited the same degree of inactivation as its susceptible counterpart in response to both chlorine and PAA disinfection. Erythromycin is part of the macrolide, lincosamide, streptogramin (MLS) group that binds to the 50S ribosomal subunit of the bacterial chromosome (Wasko et al. 2012). It interferes with the binding

of aminoacyl-tRNA to the ribosome, preventing peptide elongation, which prevents protein synthesis necessary for normal function (Schaffhauser et al. 2018). The most common erythromycin resistance gene is *ermB*, which modifies the target site of antibiotic binding, preventing the inhibition of ribosomal activity; it is associated with high levels of resistance to erythromycin (Sutcliffe et al., 1996). Previous studies have found that bacteria which inhibit ribosomal binding of antibiotics have shown slightly lower rates of chlorine inactivation, although only at concentrations greater than 1.5 mg/L free Cl₂ (Huang et al. 2013). However, the authors were not able to explain how binding site inhibition confers chlorine resistance to bacteria. Oxidation of the cell membrane by chlorine and PAA, causing lysis, would immediately cause cell death, whereas diffusion through the cell membrane and oxidation of the intercellular material would also degrade the antibiotic binding site on the 50S ribosome. There is a possibility that this previously tested strain possessed multi-drug resistance, as resistance genes are often co-selected by other antibiotics when different resistance genes are associated with the same mobile genetic element (Galvin et al. 2010). The results from this gene expression study of *ermB* in response to chlorine and PAA disinfection also support the fact that *ermB* is not involved with cross-resistance to oxidizing disinfectants. Other studies using the closely related *Enterococcus* species *E. faecium* support our finding that PAA disinfection does not result in an increase in *ermB* (Turolla et al. 2017). However, these findings should be verified in future with more studies.

4.3 Potential Mechanisms of Cross-Resistance by Meropenem-Resistant *E. fergusonii*

In contrast to the erythromycin-resistant *E. faecalis* strain, the results of this study show that the meropenem-resistant *E. fergusonii* exhibited less log inactivation in response to both chlorine and PAA, therefore this strain must possess a mechanism of resistance to oxidizing disinfectants. Meropenem is a broad-spectrum beta-lactam that is part of the carbapenem group, a class of

antibiotics that are considered to be “last resort” treatments because of their ability to inactivate a large variety of multidrug-resistant bacteria (Walsh et al. 2010). Meropenem inhibits cross-linking of transpeptides, preventing the formation of the cell wall, as well as penetrates the cell wall and binds with penicillin-binding-proteins, the most common penicillin-resistant mutation (Baldwin et al 2008). However, an emerging and dangerous class of ARGs called carbapenemases have the ability to inhibit and destroy these powerful carbapenem antibiotics (Jeon et al. 2015). Specifically, the recently discovered *bla*_{NDM-1} gene has shown that pathogens continue to evolve new and powerful antibiotic resistance mechanisms to carbapenems across the globe (Khan et al. 2017). The *bla*_{NDM-1} gene produces NDM-1, a carbapenemase enzyme that performs extracellular digestion of antibiotics through hydrolysis via two zinc ions within its enzymatic active site (Sully et al. 2016). It is possible that the zinc ions in the active site of NDM-1 have the ability to complex with hypochlorite, as seen in heme haloperoxidase enzymes, where hypochlorite complexes with the ferrous active site, and which protects the cell under oxidative stress (Raven and Dunford 2015). Other studies have shown that zinc may also complex with acetic acid under physiological conditions (Krezel and Maret 2016, Hacht 2008). Therefore, it is possible that the action of NDM-1 may reduce the effects of disinfection through complexation, since no difference in chlorine demand was found between the resistant and susceptible strains.

4.4 Draft Genome (Bacterial Chromosomal and Plasmid) Analysis Elucidates Other Potential Mechanisms of Cross-Resistance

Heightened gene expression of *bla*_{NDM-1} in response to chlorine supports the hypothesis of cross-resistance of meropenem-resistant *E. fergusonii* via the production of the NDM-1 enzyme. However, the gene expression of *bla*_{NDM-1} in response to PAA did not

significantly differ from the control, although meropenem-resistant *E. fergusonii* showed less log inactivation to PAA than its susceptible counterpart. It was hypothesized that the meropenem-resistant *E. fergusonii* strain then must possess another cross-resistance mechanism, such as peroxide scavenger genes, oxidative stress resistance genes, a thickened cell wall, or multidrug efflux pump systems (Prasad and Grobelak 2020, Wong et al. 2018, Chiang et al. 2012). In order to investigate these other potential mechanisms of cross-resistance, WGS of the resistant and susceptible *E. fergusonii* strains was performed. Gene prediction and annotation using PROKKA revealed that both strains contained peroxide scavenger *katE*, *katG*, *ahpF*, and *ahpC* genes. Both strains also contained the oxidative stress response systems *soxR* and *soxS*. Resistance due to a thickened cell wall is most often attributed to gram-positive bacteria. However, studies have shown that some bacteria can also increase the thickness of their cell wall in response to certain chlorine-containing antibiotics such as vancomycin (Monteiro et al. 2017). The results of the WGS analysis reveal that neither strain contained any vancomycin-resistance genes, which might contribute to cross-resistance to oxidizing disinfectants. However, increased cell wall strength has been attributed to bacteria that resist beta-lactam antibiotics, therefore meropenem-resistant *E. fergusonii* may resist oxidative disinfection by a strengthened cell wall.

Multidrug efflux pumps have been strongly implicated in oxidizing disinfectant cross resistance (Blanco et al. 2016). However, in this study, WGS revealed that both strains contained over 40 efflux pump-related ARGs. The meropenem-resistant *E. fergusonii* only contained 3 additional efflux pump genes, and it is unknown which of these efflux pump systems may have been actively expressing during treatment with oxidizing disinfectants. It was hypothesized that the genome of the resistant *E. fergusonii* would show a much larger percentage of efflux pumps relative to the susceptible strain; however, since both contained a large number of

efflux pumps it is unknown if the resistant strain was actively expressing those efflux pumps more thoroughly. The complex interactions of regulators involved in the expression of efflux pump systems for heavy metal and antibiotic cross-resistance has been heavily studied (Pal et al. 2017). Less is known about their role in resisting oxidizing disinfectants, as oxidation causes indiscriminate damage to efflux pump systems (Jia et al. 2015). Transcriptome sequencing may be performed in future research to elucidate the activity of these systems and requires its own special study.

In this study, WGS revealed that the meropenem-resistant *E. fergusonii* not only contained *bla*_{NDM-1}, but also *bla*_{CTX-M-15} and *bla*_{TEM-1}, both of which also work via the hydrolysis of a broad spectrum of antibiotics through the production of extracellular enzymes with functional active site (Canton et al. 2012, Palzkill 2018). The other seven ARGs found within the meropenem-resistant *E. fergusonii* were related to target site alteration for specific antibiotics, and therefore do not possess the potential for a cross-resistance mechanism. This study suggests that the production of an ESBL-related extracellular enzyme may provide cross-resistance for meropenem-resistant bacteria to both antibiotics and chlorine disinfection. Although the gene expression of NDM-1 was down regulated by PAA, these additional ESBLs and inactivating ARGs may have been responsible for the difference in inactivation by PAA. The conclusion that there is potential for ESBL-containing bacteria to resist oxidizing disinfection through inactivation or complexation with a functional ARG enzyme is the first of its kind and will require future research.

4.5 Conclusions

This study compared the disinfection kinetics of two commonly found antibiotic-resistant pathogens in the environment. These bacteria were chosen in order to compare two disparate modes of phenotypic resistance to unrelated classes of antibiotics, that do not

involve the use of efflux pumps. In addition, two different disinfectants were examined, the universally used sodium hypochlorite, also known as bleach chlorine, and the more environmentally friendly, yet less thoroughly regulated oxidizing agent known as peracetic acid. Through extensive deductive research, including gene expression experiments and whole genome sequencing, we demonstrated why only some ARBs and ARGs may increase in relative abundance during wastewater and drinking water disinfection, proving that not all ARGs provide bacteria with resistance to oxidizing disinfectants such as chlorine and PAA. This study shows that NDM-1-producing *E. fergusonii* could be potentially more dangerous than previously thought, not only exhibiting resistance to the strongest antibiotics, but to traditional wastewater and drinking water disinfection as well. While *ermB* carrying *E. faecalis* still poses a threat to human health through contamination and infection, previously studied disinfection kinetics of *Enterococci* in response to chlorine and PAA disinfection may be applied to both resistant and susceptible strains in the design of disinfection treatment systems (Turolla et al. 2017). However, in order to inactivate *bla*_{NDM-1} carrying *E. fergusonii*, greater C_T values for chlorine and PAA must be used, which increases the risk associated with these antibiotic-resistant waterborne pathogens.

Studies on the inactivation of ARBs and ARGs in wastewater and drinking water treatment are increasing, but the application of these discoveries are minimal in both the United States and abroad. Although PAA has been proposed as a stronger green alternative to chlorination, both have been shown to increase the relative abundance of certain ARBs and ARGs in wastewater and drinking water (Lin et al. 2016, Jia et al. 2015). Despite other alternatives, chlorination continues to be the most widespread disinfectant used across the world, not only in water treatment facilities (EPA Chlorine Disinfection 1999). It is paramount that the proper doses of chlorine be given to inactivate pathogens, especially those that could potentially be antibiotic-resistant. This is

especially important in low- and middle-income countries that heavily rely on chlorine disinfection, but where unregulated use of antibiotics, industrial dumping, and contaminated surface waters increase the threat of antibiotic-resistant waterborne disease (Owoseni et al. 2018). The results of this research will inform dose response control of antibiotic resistant pathogens in water and wastewater treatment facilities, as well as illuminate a potential mechanism of cross-resistance behind drinking water and wastewater disinfection.

5.0 Declaration of interest

There are no conflicts of interest to declare.

6.0 Acknowledgements

This study was supported by the United States Agency for International Development (USAID) Grant # AID-OAA-A-11-00012. The opinions and views expressed are those of authors and not necessarily reflect any association with the funding agency. We appreciate Dr. Windy Tanner, currently at Yale University, for providing us with the meropenem-resistant *E. fergusonii* strain when Dr. Tanner was at the University of Utah, and Dr. Jennifer Weidhaas at the University of Utah for providing us with the erythromycin-susceptible *E. faecalis* and *E. fergusonii* strains.

7.0 Data submission

Accession number(s): All DNA sequences for *Escherichia fergusonii* strain S1 (BioProject PRJNA700874, BioSample SAMN17843116, SRA SRR13668360) and *Escherichia fergusonii* strain S2 (BioProject PRJNA701008, BioSample SAMN17843115, SRA SRR13668385) have been submitted to GenBank and assigned accession numbers SUB9052937 (CP070279) and SUB9053365 (CP070280), respectively. CheckM analysis and PROKKA annotation files are available on github (<https://github.com/asbhattacharjee/Escherichia-fergusonii-Whole-Genome-Results>)

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Table 1. Chlorine and PAA disinfection kinetics values.

Table 1: Chlorine and PAA disinfection constant values.

Kinetics	Resistant <i>E. faecalis</i>	Susceptible <i>E. faecalis</i>	Resistant <i>E. coli</i>	Susceptible <i>E. coli</i>
Chlorine K_S (L/mg*min)	-0.239	-0.230	-0.281	-0.374
P-value	3.20×10^{-15}	8.77×10^{-14}	7.02×10^{-12}	2.00×10^{-16}
PAA K_S (L/mg*min)	-0.969	-0.874	-0.914	-1.31
P-value	3.11×10^{-16}	2.27×10^{-10}	5.14×10^{-15}	2.00×10^{-16}

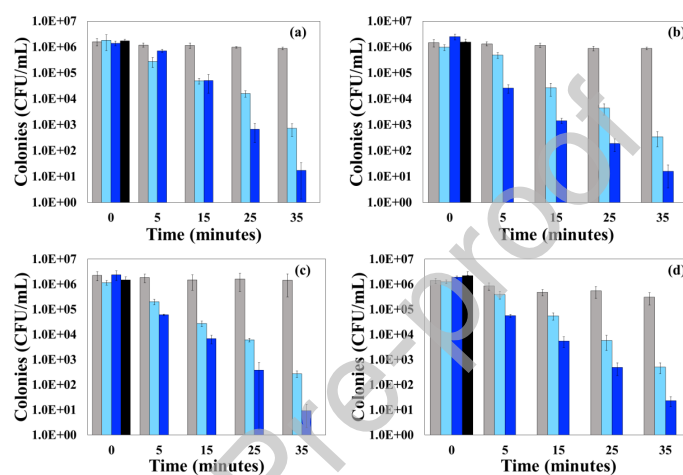
Table 2: Summary of sequenced genomes of resistant *E. fergusonii* (S1) and susceptible *E. fergusonii* (S2).

Sequencing and Assembly Metrics	<i>E. fergusonii</i> Resistant (S1)	<i>E. fergusonii</i> Susceptible (S2)
Reads	1,600,266	1,298,594
N50 Score (bp)	93,450	93,678
Coverage (%)	60.16	68.39
Plasmids	1	1
Heterogeneity (%)	0%	0%
PROKKA ARGs	65	52
CARD ARGs	74	61
Plasmid ARGs	2	0

Table 3: Statistics of Assembled Genomes of resistant *E. fergusonii* (S1) and susceptible *E. fergusonii* (S2).

Bin Names	Marker Lineage	Completeness [^] (%)	Contamination [^] (%)	Chromosome Size (bp)	Coding Density (Chr [*])	# contigs [^]	# Plasmids (bps)	# unique [^] marker s	GC [^] (%)	Predicted genes Chr [*] /Pl ^{**}	# tRNA/ tmRNA	#rRNA (5S/16S/23S)
<i>E. fergusonii</i> strain S1	Enterobacteriaceae (UID5124)	99.96	0.24	5,314,926	0.8793	125	1(55, 286)	43	50.52	5053/57	81/1	2/1/1
<i>E. fergusonii</i> strain S2	Enterobacteriaceae (UID5124)	99.81	0.04	4,608,499	0.8809	82	1(49, 476)	43	50.70	4398/56	75/1	3/1/1

[^] Bacterial chromosome statistics^{*} Chr: Chromosome^{**} Pl: Plasmid



Figures 1a-d. Chlorine inactivation for resistant and susceptible strains. (a) Resistant *E. fergusonii* (b) Susceptible *E. fergusonii* (c) Resistant *E. faecalis* (d) Susceptible *E. faecalis*. ■ 2 mg/L, ■ 4 mg/L, ■ 8 mg/L, ■ 16 mg/L

Figure 1. Chlorine inactivation for resistant and susceptible strains. (a) Resistant *E. fergusonii* (b) Susceptible *E. fergusonii* (c) Resistant *E. faecalis* (d) Susceptible *E. faecalis*

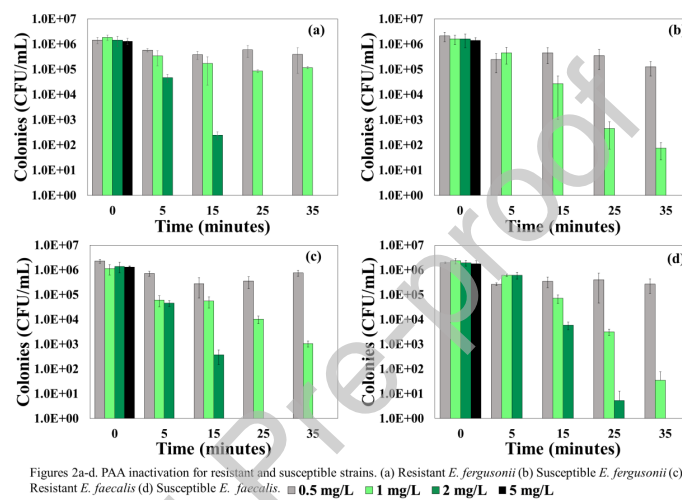
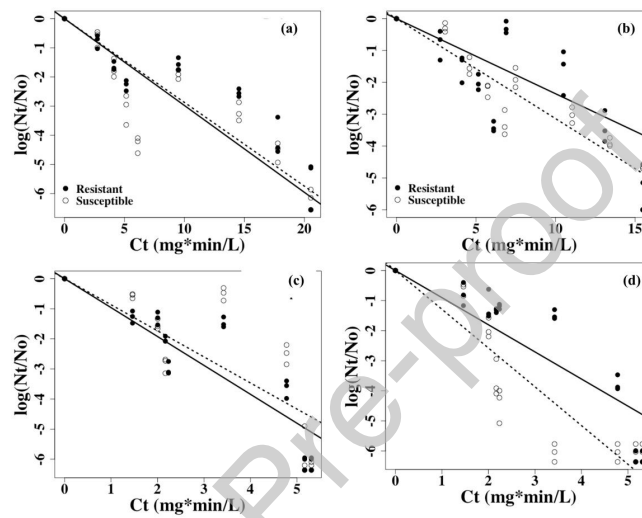
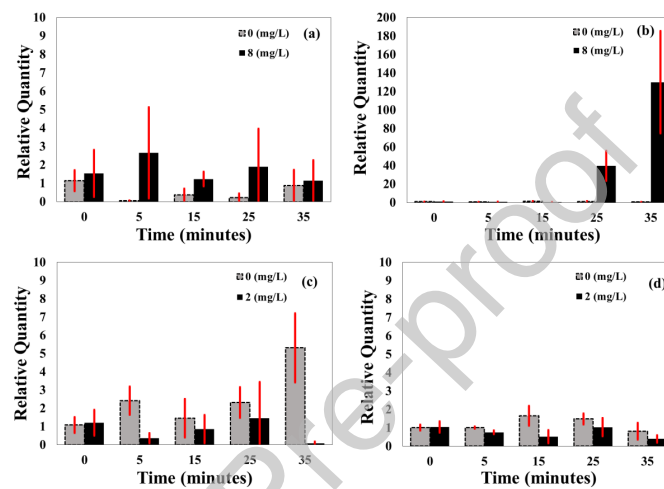


Figure 2. PAA inactivation for resistant and susceptible strains. (a) Resistant *E. fergusonii* (b) Susceptible *E. fergusonii* (c) Resistant *E. faecalis* (d) Susceptible *E. faecalis*



Figures 3a-d. ANCOVA analysis of chlorine and PAA disinfection kinetics for resistant and susceptible strains: (a) *E. faecalis* - chlorine (b) *E. fergusonii* - chlorine (c) *E. faecalis* - PAA (d) *E. fergusonii* - PAA.

Figure 3. ANCOVA analysis of chlorine and PAA disinfection kinetics for resistant and susceptible strains. (a) *E. faecalis* - chlorine (b) *E. fergusonii* - chlorine (c) *E. faecalis* - PAA (d) *E. fergusonii* - PAA.



Figures 4a-d. Gene expression of resistant strains in response to disinfection: (a) *ermB* in *E. faecalis* - chlorine (b) *bla*_{NDM-1} in *E. fergusonii* - chlorine (c) *ermB* in *E. faecalis* - PAA (d) *bla*_{NDM-1} in *E. fergusonii* - PAA.

Figure 4. Gene expression of resistant and susceptible strains in response to chlorine and PAA disinfection. (a) *ermB* in *E. faecalis* - chlorine (b) *bla*_{NDM-1} in *E. fergusonii* - chlorine (c) *ermB* in *E. faecalis* - PAA (d) *bla*_{NDM-1} in *E. fergusonii* - PAA.

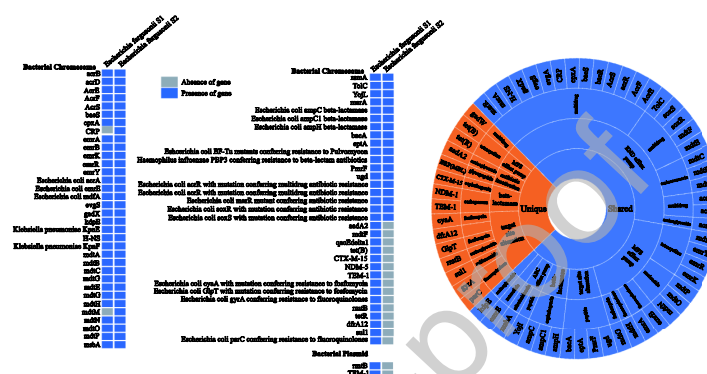


Figure 5. Presence and absence of resistance gene in *Escherichia fergusonii* chromosome and plasmid of strains S1 and S2. A Comprehensive Antibiotic Resistance Database (CARD) reference gene identifier (RGI) identified 65 and 52 resistance genes on chromosome of *Escherichia fergusonii* S1 and *Escherichia fergusonii* S2, respectively. Two resistance genes were identified on the plasmid of *Escherichia fergusonii* S1.

Figure 5. Presence and absence of resistance gene in *Escherichia fergusonii* chromosome and plasmid of strains S1 and S2. A Comprehensive Antibiotic Resistance Database (CARD) (reference) reference gene identifier (RGI) identified 65 and 52 resistance genes on chromosome of *Escherichia fergusonii* S1 and *Escherichia fergusonii* S2, respectively. Two resistance genes were identified on the plasmid of *Escherichia fergusonii* S1.

Graphical abstract

